# Case Study: Networking a Lab with 90 Systems

## Introduction:

ABC Labs is a research facility that aims to connect and network 90 computer systems within its premises. The lab requires a robust and efficient network infrastructure to support various scientific experiments, data analysis, and collaboration among researchers. This case study explores the planning phase of networking the lab and identifies potential issues that need to be addressed for successful implementation.

## 1. **Network Design and Architecture:**

The first step in planning the lab's network involves designing an appropriate network architecture. Key considerations include:

a) **Scalability:** Ensuring the network design can accommodate future expansion and addition of more systems or devices.

b) **Redundancy:** Implementing redundant network components to minimize downtime and ensure high availability.

c) **Segmentation:** Determining network segmentation to isolate sensitive data and systems from the rest of the lab.

d) **Bandwidth Requirements:** Assessing the bandwidth needs of various applications and users to allocate sufficient network resources.

2. **Infrastructure Requirements:**

Building a robust network infrastructure is crucial for the lab's operations. Some potential issues in this area include:

a) **Cabling:** Identifying the most suitable cabling infrastructure to support the network, considering factors such as cable length limitations, cable types (e.g., Ethernet, fiber optic), and cable management.

b) **Network Equipment:** Selecting and procuring the appropriate network devices, including switches, routers, and firewalls, to meet the lab's requirements in terms of performance, security, and scalability.

c) **Power and Cooling:** Ensuring adequate power supply and cooling mechanisms for the network equipment to prevent overheating or power-related failures.

d) **Physical Space**: Assessing the lab's physical layout and available space to determine the optimal placement of networking equipment and cabling routes.

3. **IP Addressing and Subnetting:**

IP addressing and subnetting play a crucial role in efficiently managing the lab's network. Potential issues may include:

a) **IP Address Exhaustion:** Ensuring the lab has enough IP addresses to accommodate all systems and devices, both present and future, by selecting an appropriate IP address range and subnetting scheme.

b) **IP Conflict Resolution:** Implementing mechanisms to handle and resolve IP address conflicts that may arise due to misconfiguration or duplication.

c) **DHCP (Dynamic Host Configuration Protocol):** Planning and configuring DHCP servers to automate IP address assignment and manage network resources effectively.

4. **Network Security:**

Protecting the lab's network and data from unauthorized access or breaches is vital. Potential issues related to network security include:

a) **Access Control:** Implementing user authentication mechanisms, such as usernames and passwords or more advanced methods like two-factor authentication, to control access to the lab's systems and resources.

b) **Firewalls and Intrusion Detection/Prevention Systems (IDS/IPS):** Deploying firewalls and IDS/IPS solutions to monitor and protect the network from malicious activities and unauthorized access attempts.

c) **Data Encryption:** Identifying the need for encryption protocols, such as VPN (Virtual Private Network) or SSL/TLS (Secure Sockets Layer/Transport Layer Security), to secure data transmission between systems.

d) **Security Auditing and Monitoring:** Establishing procedures for regular security audits and implementing network monitoring tools to detect and respond to potential security breaches.

## Conclusion:

Networking a lab with 90 systems requires meticulous planning to ensure a reliable, scalable, and secure network infrastructure. The identified issues in planning include network design and architecture, infrastructure requirements, IP addressing and subnetting, and network security. By addressing these issues during the planning phase, ABC Labs can lay a strong foundation for an efficient and robust network that meets their research and collaboration needs.